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SUPERSEDING
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MILITARY SPECIFICATION

ENAMEL, BAKING, PHENOL- OR UREA-FORMALDEHYDE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for phenol- or urea-formaldehyde, semigloss, baking enamel.

1.2 Classification. Semigloss enamel covered by this specification shall be of the following types, as specified (see 6.2):

Type I - Phenol-formaldehyde resin base
Type II - Urea-formaldehyde resin base

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
L-T-90 - Tape, Pressure-Sensitive, Adhesive (Cellophane and Cellulose Acetate)
TT-P-143 - Paint, Varnish, Lacquer, and Related Materials, Packaging, Packing and Marking of
TT-T-291 - Thinner, Paint, Volatile Spirits (Petroleum Spirits)

STANDARDS

FEDERAL
Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing
Fed. Std. No. 595 - Colors

FSC 8010

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The manufacturer is given latitude in the selection of raw materials and processes of manufacture, provided the product meets all requirements of this specification. The product shall contain no benzol (benzene), chlorinated or other toxic compounds, or hydrolyzable chlorine derivatives. The enamel and thinner shall be furnished in quantities proportional to the ratio recommended by the manufacturer to produce the required film thickness after baking.

3.1.1 Vehicle. The isolated vehicle for type I shall contain a phenol-formaldehyde resin. The isolated vehicle for type II shall contain a urea-formaldehyde resin. The vehicle shall show a negative test for rosin.

3.2 Color. The enamels shall be furnished in the color as specified in the contract or order, and of a shade conforming to Fed. Std. No. 595.

3.3 Condition in container. A freshly opened full container of the enamel shall show no more pigment settling or caking than can be easily and completely reincorporated to a smooth homogeneous state when tested as in 4.5.3.

3.4 Storage stability.

3.4.1 Full container. The enamel shall show no curdling, livering, caking, or color separation and shall be free from lumps and skins for at least six (6) months after delivery by the manufacturer to the Government (see 4.5.4.1).

3.4.2 Partially full container. After 30 days storage, a 3/4 filled 8-ounce closed glass jar of the enamel shall show no livering, curdling, color separation or hard caking when tested as in 4.5.4.2. The enamel shall mix readily to a smooth homogeneous state, and any skin formed shall be continuous and easily removed.

3.5 Spraying properties. The enamel, when tested as in 4.5.5 shall spray satisfactorily in all respects and shall level out to a smooth film free from pebbling or other surface imperfections.

3.6 Coarse particles and skins. Coarse particles and skins shall not exceed 1 percent by weight of pigment when tested as in 4.5.6.

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3.7 Fineness of grind. The fineness of grind shall be a minimum of 6 when tested as in 4.5.7.

3.8 Odor. The enamel shall not have an objectionable odor when tested as in 4.5.8.

3.9 Drying time. The enamel shall be dry through after baking 45 minutes at 375°F, or in accordance with a time-temperature cycle recommended by the manufacturer and approved by the contracting officer, when tested as in 4.5.9.

3.10 Gloss. The enamel shall have a 60 degree specular gloss of 25 to 40 when tested as in 4.5.10.

3.11 Flexibility. A film of the enamel shall show no cracking, chipping or lifting when tested as in 4.5.11.

3.12 Adhesion. A film of the enamel when tested as in 4.5.12, shall show no removal of the coating by the adhesive tape.

3.13 Autoclave test. There shall be no softening or change in appearance or flexibility of the film of enamel when tested as in 4.5.13.

3.14 Immersion resistance.

3.14.1 Cold water resistance. A film of the enamel, when tested as in 4.5.14.1, shall show no wrinkling or blistering immediately after removal of the panel from the water. The enamel shall be no more than slightly affected when examined 2 hours after removal. After 24 hours air drying, the portion of the panel which was immersed shall be almost indistinguishable with regard to hardness, color and gloss from the portion which was not immersed.

3.14.2 Hot water resistance. A film of the enamel, when tested as in 4.5.14.2, shall show no wrinkling or blistering immediately after removal of the panel from the boiling water. The enamel shall be no more than slightly affected when examined 2 hours after removal. After 24 hours air drying, the portion of the panel which was immersed shall be almost indistinguishable with regard to hardness, color and gloss from the portion which was not immersed.

3.14.3 Mineral spirits resistance. A film of the enamel, when tested as in 4.5.14.3, shall show no wrinkling or blistering immediately after removal of the panel from the mineral spirits. The enamel shall be no more than slightly affected when examined 2 hours after removal. After 24 hours air drying, the portion of the panel which was immersed shall be almost indistinguishable with regard to hardness, color and gloss from the portion which was not immersed.

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3.14.4 Sodium hydroxide resistance. A film of the enamel, when tested as in 4.5.14.4, shall not have softened or otherwise deteriorated immediately after removal of the panel from the sodium hydroxide solution.

3.14.5 Chlorine resistance. When tested as in 4.5.14.5, a film of the enamel shall show no visible effects of the immersion in chlorine solution immediately after removal nor shall the panel show any rusting.

3.15 Accelerated weathering. A film of enamel tested as in 4.5.15 shall show no more than light chalking (No. 6, method 6411 of Fed. Test Method Std. No. 141) and a color change not exceeding 6 units.

3.16 Abrasion resistance. The average weight of enamel coating removed shall not exceed 18 mg when tested as indicated in 4.5.16.

3.17 Salt spray resistance. A film of enamel, tested as in 4.5.17 and examined immediately after removal from the test, shall show no rust creepage beyond one-eighth inch from the score. The enamel shall show no wrinkling or blistering. After 24 hours air drying, the enamel shall be no softer than an unexposed film of the enamel prepared in the same manner.

3.18 Workmanship. The ingredients of the enamel shall be intimately compounded and processed to produce a uniform product that meets all the requirements of the specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Size of lot. A lot shall consist of enamel from one batch submitted at one time for inspection.

4.3 Sampling and inspection. Sampling and inspection shall be performed in accordance with section 1000 of Fed. Test Method Std. No. 141.

4.4 Classification of tests. Testing under this specification shall be for the purpose of acceptance of lots.

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4.5 Test methods.

4.5.1 Vehicle. The vehicle shall be isolated by method 4032 of Fed. Test Method Std. No. 141. Suitable qualitative tests shall then be run using classical techniques, infrared, or pyrolysis-gas chromatography to determine compliance with the requirements in paragraph 3.1.1. The test for rosin shall be conducted in accordance with method 5031 of Fed. Test Method Std. No. 141.

4.5.2 Color. Compare a dried film of the test enamel with the appropriate color standard of Fed. Std. No. 595, for compliance with 3.2.

4.5.3 Condition in container. Determine package condition in accordance with method 3011 of Fed. Test Method Std. No. 141. Observe for compliance with 3.3.

4.5.4 Storage stability.

4.5.4.1 Full container. The Government at its option and at any time not to exceed six (6) months after delivery, may test the enamel stored in its original containers in accordance with method 3022 of Fed. Test Method Std. No. 141 for compliance with 3.4.1.

4.5.4.2 Partially full container. In accordance with method 3021 of Fed. Test Method Std. No. 141, allow a 3/4 full 8-ounce glass jar of the enamel to stand undisturbed for 30 days and examine for compliance with 3.4.2.

4.5.5 Spraying properties. Reduce the enamel in accordance with the manufacturer's instructions. Test in accordance with method 4331 of Fed. Test Method Std. No. 141 using a distance of 8 to 10 inches from test panel to gun. Observe for spraying properties of method 4331 and check for compliance with 3.5.

4.5.6 Coarse particles and skins. Determine the weight of coarse particles and skins in accordance with method 4092 of Fed. Test Method Std. No. 141. Check for compliance with 3.6 using pigment content as determined by method 4021 of Fed. Test Method Std. No. 141.

4.5.7 Fineness of grind. Fineness of grind shall be determined in accordance with method 4411 of Fed. Test Method Std. No. 141.

4.5.8 Odor. The test for objectionable odor shall be conducted in accordance with method 4401 of Fed. Test Method Std. No. 141.

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4.5.9 Drying time.

4.5.9.1 Preparation of test panels. Unless otherwise specified in the contract or order, use 4 by 6 inch steel panels which have been solvent cleaned and phosphoric acid etched in accordance with procedure B, method 2011 of Fed. Test Method Std. No. 141. The enamel, reduced in accordance with the manufacturer's instructions, shall be sprayed to give a dry film thickness of 0.0009 to 0.0011 inch. The panels shall be baked for 45 minutes at 375°F and then air dried for 24 hours prior to testing.

4.5.9.2 Test procedure. Determine dry through time in accordance with method 4061 of Fed. Test Method Std. No. 141. Check for compliance with 3.9.

4.5.10 Gloss. The 60° specular gloss shall be determined in accordance with method 6101 of Fed. Test Method Std. No. 141.

4.5.11 Flexibility. Six panels shall be prepared as in 4.5.9.1 except that the panels shall be 31 gage steel. The flexibility test shall be conducted in accordance with method 6221 of Fed. Test Method Std. No. 141 using a 1/8 inch mandrel. Examine for compliance with 3.11.

4.5.12 Adhesion. Pressure-sensitive tape conforming to L-T-90 shall be applied to the bend of the coated panels used in the flexibility test above (4.5.11). The tape shall be pressed down firmly and then quickly stripped off. Check for compliance with 3.12.

4.5.13 Autoclave test. Two steel panels prepared as in 4.5.9.1 shall be exposed to saturated steam at a pressure of 17 pounds per square inch for 3 minutes. After exposure to steam the flexibility test of 4.5.11 shall be conducted on the panels. Check for compliance with 3.13.

4.5.14 Immersion resistance.

4.5.14.1 Cold water resistance. Two steel panels prepared as in 4.5.9.1 shall be immersed in distilled water at 77 ± 2°F (25 ± 1°C) for 24 hours in accordance with method 6011 of Fed. Test Method Std. No. 141. Check for compliance with 3.14.1.

4.5.14.2 Hot water resistance. Two steel panels prepared as in 4.5.9.1 shall be immersed in boiling distilled water for 15 minutes in accordance with method 6011 of Fed. Test Method Std. No. 141. Check for compliance with 3.14.2.

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4.5.14.3 Mineral spirits resistance. Two steel panels prepared as in 4.5.9.1 shall be immersed in mineral spirits conforming to TT-T-291 at 77 ± 2°F (25 ± 1°C) for 16 hours in accordance with method 6011 of Fed. Test Method Std. No. 141. Check for compliance with 3.14.3.

4.5.14.4 Sodium hydroxide resistance. Two steel panels prepared as in 4.5.9.1 shall be immersed in 0.1N sodium hydroxide (NaOH) solution at 77 ± 2°F (25 ± 1°C) for 48 hours in accordance with method 6011 of Fed. Test Method Std. No. 141. Check for compliance with 3.14.4.

4.5.14.5 Chlorine resistance. Two steel panels prepared as in 4.5.9.1 shall be immersed in 0.1 percent available chlorine solution at 77 ± 2°F (25 ± 1°C) for 48 hours in accordance with method 6011 of Fed. Test Method Std. No. 141. (The solution shall be prepared by the appropriate dilution of commercial sodium hypochloride solution.) Check for compliance with 3.14.5.

4.5.15 Accelerated weathering. Prepare four steel panels as in 4.5.9.1. Measure the color as in method 6123 of Fed. Test Method Std. No. 141. Expose the panels for 150 hours to accelerated weathering in accordance with method 6151 of Fed. Test Method Std. No. 141. Examine the exposed panels for chalking by method 6411 of Fed. Test Method Std. No. 141. Again measure the color and determine the color difference. Check for compliance with 3.15.

4.5.16 Abrasion resistance. Three steel panels shall be prepared as in 4.5.9.1 except that the panels should be 4 inches by 4 inches for use on a Taber Abraser. Weigh each coated panel and record the weight in milligrams. Test for abrasion resistance in accordance with method 6192 of Fed. Test Method Std. No. 141 using CS-10 calibrase wheels and 1000 grams weight on each wheel. Five hundred (500) wear cycles shall be run on each panel. Reweigh the panels, calculate the loss in weight and check for compliance with 3.16.

4.5.17 Salt spray resistance. Two steel panels shall be prepared as in 4.5.9.1. Score and expose to 5 percent salt spray for 250 hours in accordance with method 6061 of Fed. Test Method Std. No. 141. Examine for compliance with 3.17.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. The enamel shall be packaged, packed and marked in accordance with TT-P-143. The level of packaging shall be A or C and the level of packing shall be A, B, or C as specified (see 6.2). The enamel shall be furnished in 1 gallon multiple friction top containers, in 5 gallon lug cover steel pails or in 55 gallon steel drums, as specified (see 6.2).

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6. NOTES

6.1 Intended use. The enamel covered by this specification is intended for the Sabot for 105mm, M392, projectiles. The phenol-formaldehyde coating is intended primarily for use over steel and the urea-formaldehyde over zinc coated steel (personnel hardware such as buckles, hooks, snaps, etc.).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required.
(c) Size of container required (see section 5).
(d) Level of packaging and packing required (see section 5).
(e) Color required. Specify color number from Fed. Std. No. 595 (see 3.2).
(f) If panel metal other than steel is to be used for testing it shall be stated in the contract.

Custodians:
Army - MR
Navy - SH

Preparing activity:
Army - MR

Project No. 8010-0612

Review activities:
Army - GL, MR, MU
DSA - IS

User activities:
Army - MU
Navy - SH

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